

The Imprint of Star Formation Seen on Parsec-scale Molecular Gas Profiles in NGC 253

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ABSTRACT

The internal density structure of molecular clouds governs where, when, and how efficiently stars form, but has rarely been measured directly on few-parsec scales. Here, we present radial intensity profiles of 215 resolved molecular gas peaks in the nearby star-forming spiral NGC 253. The profiles are measured from ALMA ¹²CO(2–1) observations at 7 pc resolution across a 1.4×5.6 kpc field in the disk. Stacking the background-subtracted profiles reveals a characteristic two-component form. The inner profile (with density power-law index $\alpha_{\text{in}} = 1.65^{+0.02}_{-0.05}$) matches predictions for self-gravitating collapse and sub-parsec scale observations, but steepens to $\alpha_{\text{out}} \approx 2.7$ at larger radii. This steepening is largely geometric, as the stacked peaks reach an elongated aspect ratio of $q = 2.4^{+0.1}_{-0.2}$ by $r = 30$ pc, with a milder steepening along their major axes. To test how star formation and feedback reshape this structure, we compare the CO profiles to spatially matched VLT/MUSE H α and JWST/MIRI 21 μm imaging of the ionized and dusty interstellar medium (ISM), a combination previously not available beyond the Local Group at these scales. Peaks near H α and 21 μm emission are systematically brighter in CO at all radii, yet their normalized median profile shapes appear unchanged. Their degree of disruption, however, does vary: scatter, asymmetry, and elongation all increase modestly with H α association, with similar but much weaker trends for 21 μm . The imprint of star formation thus disturbs the regularity of these structures while leaving their median shape intact, with the 21 μm vs. H α contrast hinting at a sequence sampled from embedded to exposed star formation.

Keywords: Interstellar medium (847); Molecular clouds (1072); Molecular gas (1073); Star formation (1569); Stellar feedback (1602); Starburst galaxies (1570)

1. INTRODUCTION

The ability of gas to form stars depends, fundamentally, on the internal density and dynamical structure of molecular clouds (Kennicutt & Evans 2012; Krumholz 2014; Schinnerer & Leroy 2024). A cloud’s radial density profile encodes this arrangement. It is set by the balance between self-gravity and the support supplied by turbulence, magnetic fields, and thermal pressure (McKee & Ostriker 2007; Crutcher 2012; Padoan et al. 2014), as well as by the disruptive action of stellar feedback (Dale et al. 2014; Krumholz et al. 2014). The shape of

the molecular gas density profile is therefore among the most informative diagnostics of cloud physics.

To describe the density profiles within individual gas structures, theoretical models generally invoke power-law density distributions of the form $\rho(r) \propto r^{-\alpha}$ with a slope set by the dominant physics: gravitational collapse drives $\alpha \approx 1.5$ –2 (Shu 1977; Naranjo-Romero et al. 2015; Vázquez-Semadeni et al. 2019; Gómez et al. 2021), while pressure and magnetic support can flatten the profile toward $\alpha = 1$. These are typically scale-free solutions calibrated at the $\lesssim 1$ pc scales of protostellar cores, but the larger-scale shape also has direct consequences for how stars form, for example by setting the behavior of the star formation efficiency per free-fall time (Federath & Klessen 2013; Burkhardt 2018; Meidt et al. 2025; Zamora-Avilés et al. 2025). In simulated clouds, steep

profiles collapse to form massive, bound clusters, while flatter profiles tend toward distributed fragmentation with lower-mass subclusters and longer free-fall times (Miyama et al. 1987; Girichidis et al. 2011; Chen et al. 2021).

Testing these theoretical expectations requires establishing the characteristic density profile over a large and diverse population of molecular gas peaks, and comparing such structure to signatures of recent star formation. Current observations, however, are largely confined to relatively small samples of nearby cores and clumps on ~ 0.1 – 1 pc scales in the Milky Way. These data show that density profiles broadly follow $\alpha \approx 1.5$ – 2 (see review by Beuther et al. 2025), though their behavior at larger radii above the core remains contested across different samples (e.g., Pirogov 2009; Kauffmann et al. 2010; Peretto et al. 2023; Beuther et al. 2024). Profiles may also evolve over time, steepening as gas initially collapses (Beuther et al. 2002; Butler & Tan 2012; Caldwell & Chang 2018; Li & Zhou 2022) before feedback disrupts them and reduces the central gas concentration (Chevance et al. 2020; Chen et al. 2021; Gieser et al. 2023).

Extragalactic surveys such as PHANGS-ALMA (Leroy et al. 2021a) can mitigate the distance uncertainties and modest sample sizes that hinder our view in the Milky Way. By stacking the CO emission of ~ 9000 clouds, He et al. (2026) recover a two-component radial profile at 150 pc resolution, which traces the clustering of clouds. At the few-parsec scales where individual star-forming regions assemble, a corresponding statistical analysis of cloud structure in a metal-rich target has not been attempted beyond the Milky Way.

In this paper, we conduct the first systematic extragalactic measurements of resolved cloud density profiles with associated star formation tracers, extending the local view of molecular gas peak structure to a new spatial and statistical regime. To do so, we target NGC 253, the nearest massive, actively star-forming galaxy accessible to ALMA ($D = 3.5 \pm 0.2$ Mpc, Okamoto et al. 2024). With a molecule-rich disk environment comparable in stellar mass, morphology, and star formation rate to the Milky Way (Leroy et al. 2021b; McClain et al. 2026), NGC 253 is an ideal laboratory to study molecular gas structure and the star formation cycle from an external perspective. We combine 7 pc resolution ALMA $^{12}\text{CO}(2-1)$ observations over a 1.4×5.6 kpc field (Oakes et al. 2025, hereafter O25) with spatially matched VLT/MUSE $\text{H}\alpha$ (Congiu et al. 2025) and JWST/MIRI $21\ \mu\text{m}$ (McClain et al., in prep) imaging (§2). We identify 215 CO peaks and measure their radial structure and key physical properties (§3),

then stack them to characterize the typical profile shape and test its dependence on local star formation (§4).

2. DATA: A MULTIWAVELENGTH ZOOM INTO NGC 253

We analyze the high-resolution ($\theta_{\text{beam}} = 0''.42 \approx 7.2$ pc) ALMA $^{12}\text{CO}(2-1)$ observations of molecular gas in NGC 253 (PI: E. Rosolowsky) presented by O25 (Fig. 1a). The observations cover a 1.4×5.6 kpc (de-projected) field ~ 2 – 5 kpc from the galaxy’s center, chosen as a representative region in the star-forming disk spanning spiral arms. They combine 12 m, 7 m, and total power data to recover emission on all spatial scales. The typical $1\sigma_{\text{rms}}$ noise of 450 mK per 1 km s^{-1} channel corresponds to mass surface density and point-mass sensitivities of $5.9\text{ M}_{\odot}\text{ pc}^{-2}$ and $340\text{ M}_{\odot}$ within the circularized beam.¹ These spatial and mass scales indicate that we are probing the substructure of individual clouds and approaching the scales of large clumps within them (Ballesteros-Paredes et al. 2020). Besides the emission cube itself, we also use the associated high-completeness “broad” signal mask, the strictly masked intensity-weighted mean velocity (moment 1) map, and RMS noise cube (Leroy et al. 2021a; O25).

To trace recent star formation activity, we use an extinction-corrected² $\text{H}\alpha$ map from the VLT/MUSE survey of NGC 253 (Fig. 1b; Congiu et al. 2025). We adopt the Gaussian-fit $\text{H}\alpha$ flux maps from the PHANGS-MUSE data analysis pipeline (Emsellem et al. 2022). After convolving with a $0''.6$ kernel to increase the signal-to-noise ratio, the mean spatial resolution is $1''.0 \approx 17$ pc. Over the region of our ALMA field, the MUSE $\text{H}\alpha$ intensity map has a median 1σ uncertainty of $2.6 \times 10^{-18}\text{ erg s}^{-1}\text{ cm}^{-2}\text{ arcsec}^{-2}$, corresponding to a point-source sensitivity limit of $4.3 \times 10^{33}\text{ erg s}^{-1}$ at the distance of NGC 253.

To capture the $21\ \mu\text{m}$ emission from stochastically heated small dust grains associated with young star-forming regions, we use JWST/MIRI F2100W imaging (Fig. 1c; McClain et al., in prep). These data were obtained as part of Cycle 2 program GO #2987 (PI: A. Leroy) and reduced using the PHANGS-JWST pipeline (Lee et al. 2023; Williams et al. 2024). After convolving to match the mean $1''.0$ resolution of the MUSE map, the

¹ Assuming $D = 3.5$ Mpc with a field-calibrated $\alpha_{\text{CO}}^{(1-0)} = 3.68\text{ M}_{\odot}(\text{pc}^2\text{ K km s}^{-1})^{-1}$ and $R_{21} = 0.615$ (equivalently $\alpha_{\text{CO}}^{(2-1)} = 5.99\text{ M}_{\odot}(\text{pc}^2\text{ K km s}^{-1})^{-1}$), per O25’s Appendix B. Sensitivities are quoted for a line width of 5 km s^{-1} .

² Using the O’Donnell (1994) extinction curve with an intrinsic $\text{H}\alpha/\text{H}\beta$ ratio of 2.86 and $R_V = 3.1$.

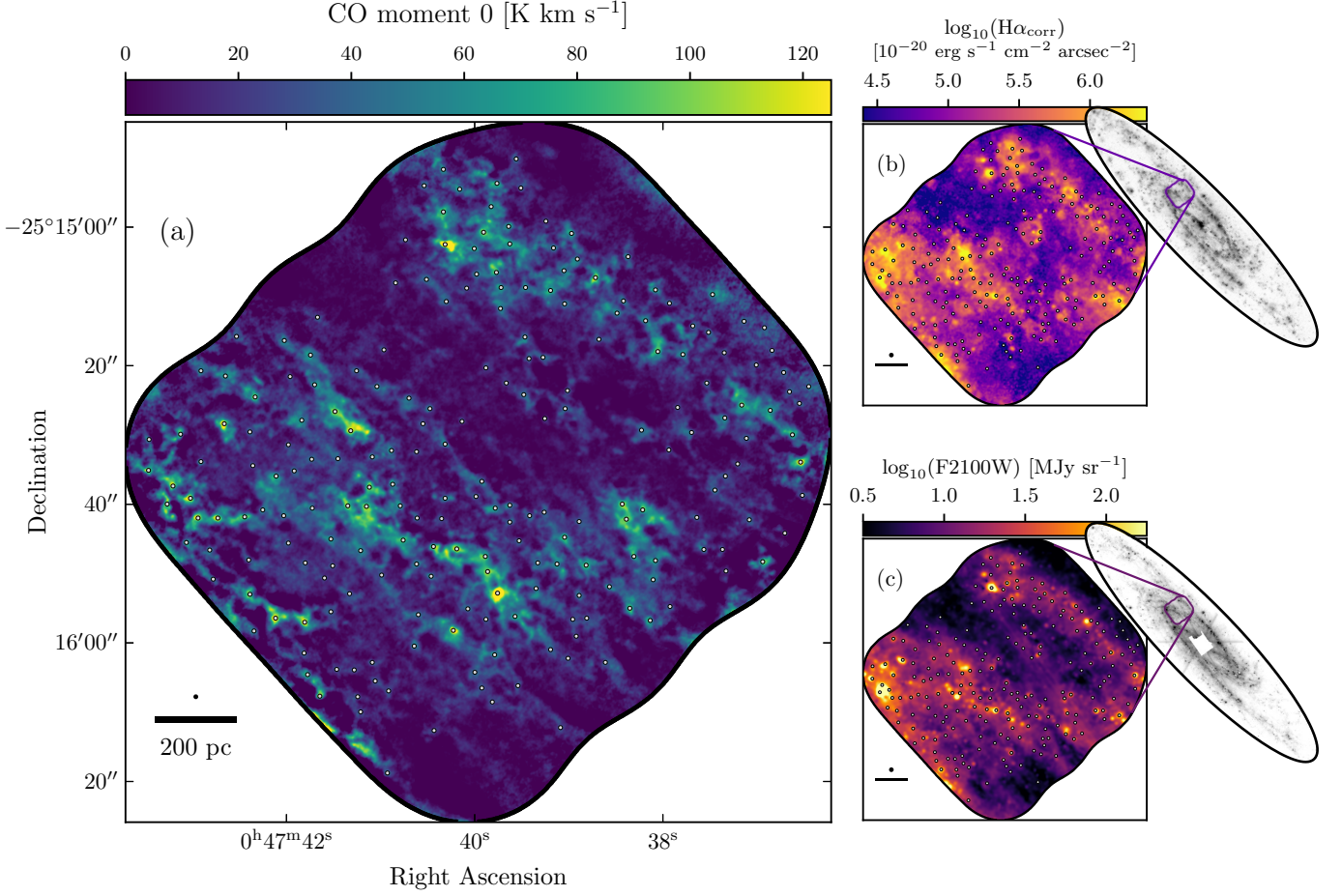


Figure 1. A multiwavelength view of our field in NGC 253, showing (a) ALMA $^{12}\text{CO}(2-1)$ integrated intensity (moment 0), (b) MUSE extinction-corrected $\text{H}\alpha$, and (c) JWST F2100W surface brightness. Panels (b) and (c) also show full-disk maps of the same tracer. The 215 CO peaks used in this work are highlighted as gray points. A 200 pc scale bar, along with the FWHM resolution, is indicated in each panel.

data achieve a median 1σ noise level of 0.08 MJy sr^{-1} over our field.

3. METHODS: PEAKS AND PROFILES

3.1. Building a spectrally masked peak catalog

We generate a catalog of bright, well-separated CO peaks from the broadly masked integrated intensity (moment 0) map. We identify local maxima via `scipy.ndimage.maximum_filter` with a circular search kernel of radius 30 pc, retaining only those brighter than $30 \text{ K km s}^{-1} \approx 180 \text{ M}_{\odot} \text{ pc}^{-2}$ ($12\sigma_{\text{mom0}}$, where $\sigma_{\text{mom0}} = 2.5 \text{ K km s}^{-1}$ is the median uncertainty of the masked maps) and farther than $4\theta_{\text{beam}} \approx 29 \text{ pc}$ from the image edge. These thresholds select prominent, independent features over extended or low-contrast emission, yielding 215 spatially distinct peaks (Fig. 1). They encompass the bulk of the molecular emission, as 50% of the total CO flux lies within 30 pc of a cataloged peak, and 88% within 60 pc.

Because NGC 253’s high inclination makes the moment 0 map susceptible to line-of-sight contamination (Appendix A), we isolate the primary velocity component of emission toward each peak. We extract the mean spectrum within a circular aperture of radius $\theta_{\text{beam}}/2 \approx 3.6 \text{ pc}$ and decompose it into Gaussian velocity components (Fig. 2, bottom) with `pyspeckit` (Ginsburg & Mirocha 2011; Ginsburg et al. 2022).³ For the 96/215 peaks with multiple significant velocity components, we designate a “primary” component whose centroid velocity lies closest to the median of the strictly masked CO moment 1 map within the aperture.⁴ As

³ Via the `scousepy` Python package (Henshaw et al. 2016, 2019), with signal-to-noise and kernel width both set to 3 except where adjusting values better captures the spectral structure.

⁴ To account for blended components, we require $|\frac{v_i - v_{\text{CO}}}{v_j - v_{\text{CO}}}| < 0.8$ and $|v_i - v_j| > 3\sqrt{\sigma_{v,i}^2 + \sigma_{v,j}^2}$. Failing this, we define a combined component with $\bar{v} = (v_i + v_j)/2$ and $\sigma_v =$

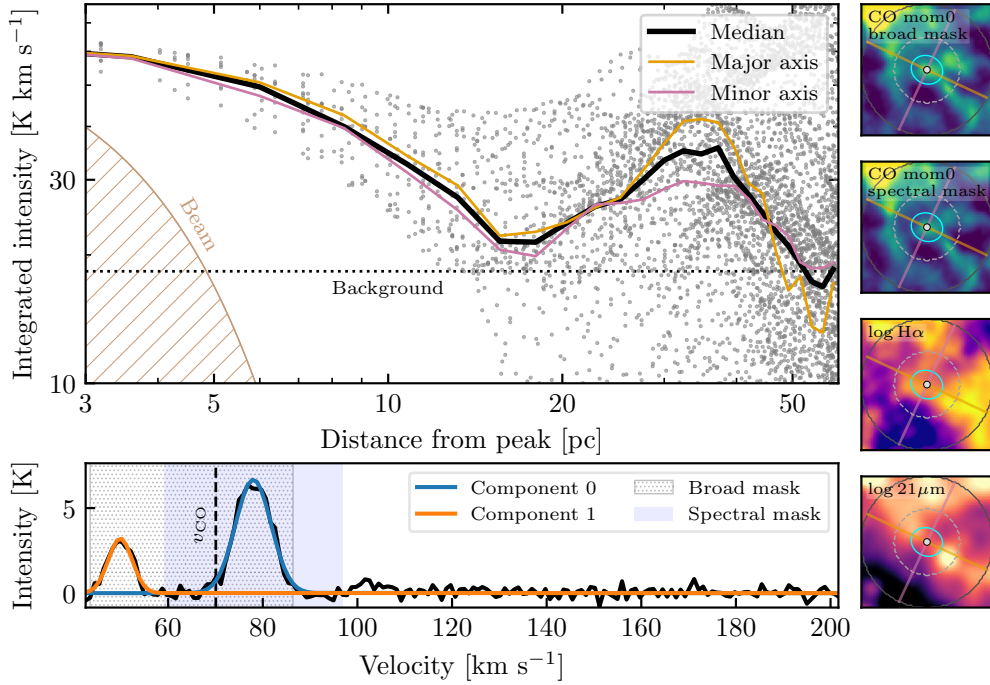


Figure 2. Main panel: Radial ^{12}CO intensity profile for peak 31. Individual pixels of the masked moment 0 map are shown as gray points, with the azimuthally averaged median profile in black. **Bottom panel:** The mean CO spectrum around $r < 3.6$ pc of the peak (black), in this case decomposed into two significant Gaussian components. Using the peak’s characteristic velocity (vertical dashed line), we select a primary component and generate a spectral mask (blue) based on its width (§3.1). **Postage stamps:** Cutouts around the peak, as labeled. The maximum radial range of 60 pc (gray) and 30 pc (white, dashed) are indicated, along with the major and minor axes of the second-moment ellipse (cyan, §3.2).

the moment 1 is intensity-weighted, this generally selects the brightest component. Multi-component incidence is uncorrelated with either star formation tracer, so this choice does not bias the trends in §4.3. We then integrate the cube over $\pm 5\sigma_v$ about this centroid, where σ_v is the component’s Gaussian width, within the broad emission mask. These spectrally filtered maps discard $>20\%$ of the broad mask CO flux for a fifth of peaks and form the basis for all subsequent analysis.

3.2. Aspect ratios and radial profiles

With spectrally filtered moment 0 maps in hand, we estimate each peak’s background over the annulus $30 < r/\text{pc} \leq 60$ as the median after iterative 2σ clipping.⁵ We then characterize each peak’s position angle and elongation from the deconvolved intensity-weighted second moments of its background-subtracted emission,

$\max(|v_i - v_j|/4, \sigma_{v,i}, \sigma_{v,j})$ and analyze this combined component.

⁵ Sigma-clipping yields more consistently convergent backgrounds than percentile or absolute thresholds; Appendix B shows that alternative background prescriptions leave our qualitative results unchanged.

following the CPROPS implementation of Rosolowsky & Leroy (2006) (ellipses in Fig. 2).

Finally, we construct azimuthally averaged radial intensity profiles in uniform bins of width $\theta_{\text{beam}}/3 \approx 2.4$ pc, recording percentiles of intensity in each bin. We use the median azimuthal profile throughout since it best suppresses contamination from sources unassociated with the peak. The main panel of Fig. 2 features a representative profile decomposed along the full azimuth as well as along major- and minor-axis quadrants, each averaging emission within 45° of the corresponding axis.

4. RESULTS: UNPACKING THE PEAK POPULATION

4.1. Overview of the CO peaks

The 215 CO peaks (§3.1) follow the two spiral arms crossing our field of view (Fig. 1), with a typical projected nearest-neighbor separation of 55_{-13}^{+17} pc.⁶ Because this separation is smaller than our 60 pc profile extent, neighboring apertures may overlap at large radii;

⁶ Throughout, statistics are quoted as the median with super/subscripts giving the 16th and 84th percentiles of the sample distribution, unless otherwise noted.

a watershed test in §4.2 demonstrates that our qualitative results are unaffected.

Peaks span central intensities of 47^{+35}_{-20} K km s⁻¹ above local sigma-clipped backgrounds of 8^{+12}_{-5} K km s⁻¹ (§3.2). Within 30 pc, they have luminosities of $(3^{+4}_{-2}) \times 10^4$ K km s⁻¹ pc², corresponding to total molecular gas masses of $(1.9^{+2.5}_{-1.4}) \times 10^5$ M_⊙. The associated spectral components have Gaussian FWHM of 9^{+5}_{-3} km s⁻¹ (§3.1). Beam-convolved Gaussian fits to individual peaks give a characteristic FWHM of 21^{+14}_{-8} pc, though we note that the profiles are not intrinsically Gaussian (§4.2). These properties — masses of $\sim 10^5$ M_⊙, velocity line widths of a few km s⁻¹, and sizes of ~ 20 pc — are comparable to well-studied Galactic clouds such as the Orion complexes (Bally 2008; Heyer & Dame 2015) and sit near the middle of the multiscale O25 hierarchy in the same field. At the 1–10% efficiencies typical of cloud-scale star formation (Chevance et al. 2023), we are in the regime of stochastically sampled associations rather than massive clusters (Ashworth et al. 2017).

4.2. Geometries of stacked profiles

We stack each radial intensity profile in Fig. 3, averaging over non-axisymmetric structure and local fluctuations to reveal the characteristic geometry. Panel (a) shows the raw stacked profiles. In panel (b) we subtract a local background from each peak (§3.2) to isolate it against more diffuse emission and to enable comparisons across the map. The median background of 8 K km s⁻¹ closely matches the median intensity > 30 pc from every peak and so is representative of the typical inter-peak gas. Profiles reach this background at 20–60 pc, with emission unrelated to the central peak driving increased scatter beyond $r \approx 30$ pc.

4.2.1. An elongated, two-component form

Panels (c) and (d) of Fig. 3 decompose the profiles along each peak’s major and minor axes (defined in §3.2). The ratio of radii at which they reach a given intensity level gives the isophotal axis ratio $q(r)$. At small radii, the circular beam drives q toward unity and we observe a round inner structure ($q = 1.15^{+0.04}_{-0.04}$ at 7 pc).⁷ Farther out, however, peaks become increasingly elongated, reaching $q = 1.8^{+0.1}_{-0.1}$ at 20 pc and $2.4^{+0.1}_{-0.2}$ at 30 pc. Beyond this point, the minor-axis profiles reach background and q is not well-defined. The same trend is present before background subtraction, pointing to substantial filamentary structure on 20–60 pc scales (also

evident in Fig. 1). The median two-dimensional stack shows that this structure is partially disk-aligned (Appendix A), consistent with in-plane projection or a preferential filamentary axis.

In panel (e) we normalize each background-subtracted profile to its central (peak) intensity. This further homogenizes the profile shapes, although significant scatter remains above the beam scale which we find to be partly systematic at the population level (§4.3). The corresponding cumulative distributions (panel h) show that the median profile encloses half its total 60 pc flux within ~ 20 pc.

Core populations at smaller scales are commonly described by a single analytic form (e.g., Beuther et al. 2025). To test whether the profiles in NGC 253 follow such a form, we fit four models to the background-subtracted, peak-normalized median profile within $r < 60$ pc: single and double Gaussian fits, as

$$I(r) = I_G e^{-r^2/(2\sigma_G^2)} \quad (1)$$

$$I(r) = I_{G,1} e^{-r^2/(2\sigma_{G,1}^2)} + I_{G,2} e^{-r^2/(2\sigma_{G,2}^2)} \quad (2)$$

and single and broken power laws, as

$$I(r) = I_p (r/r_c)^{-\gamma} \quad (3)$$

$$I(r) = \begin{cases} I_p (r/r_c)^{-\gamma_{\text{in}}}, & r \leq r_{\text{break}} \\ I_p (r_{\text{break}}/r_c)^{-\gamma_{\text{in}}} (r/r_{\text{break}})^{-\gamma_{\text{out}}}, & r > r_{\text{break}} \end{cases} \quad (4)$$

where $r_c = 10$ pc is a fixed normalization radius. Each model is evaluated in two dimensions, convolved with the synthesized beam, and averaged in the same radial bins as the data before fitting.⁸ Panels (f) and (g) compare the data to the Gaussian and power-law fits, respectively.

Table 1 lists the fit parameters, with goodness of fit quantified by the Bayesian information criterion (BIC; Schwarz 1978) and the model–data RMS deviation. Within each family, adding a second component to the fit is decisively preferred ($\Delta\text{BIC} > 40$, $\Delta\text{RMS} > 60\%$). Because the BIC already penalizes additional parameters, this indicates a real transition in the profile shape.

The profile is therefore not a monolithic structure with a single slope or width (see the robustness tests below and in Appendix B). The outer envelope must be modeled separately from the central peak, echoing the two-component structure found by He et al. (2026) for CO structures at 150 pc scales and the HII region morphology described by McClain et al. (2026) in this galaxy.

⁷ Where uncertainties follow a bootstrap over the peak sample, resampling with replacement.

⁸ We fit by unweighted least-squares minimization with `scipy.optimize.curve_fit` (Virtanen et al. 2020).

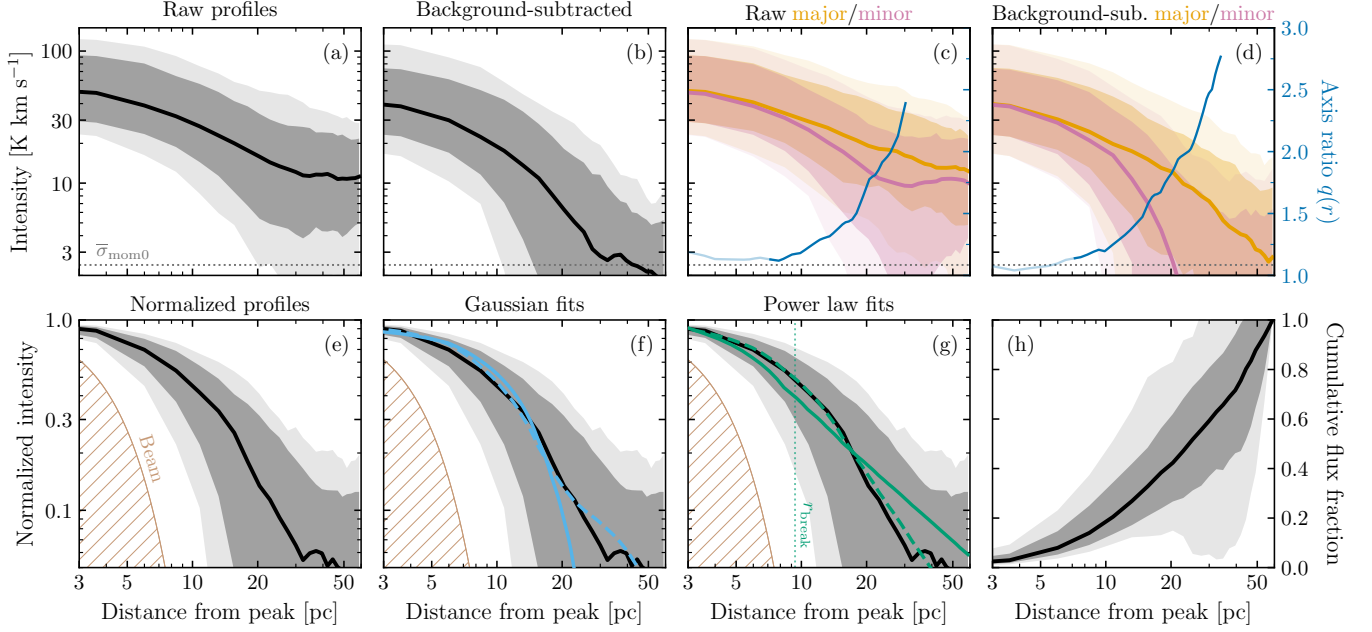


Figure 3. Stacked radial profiles of the CO peak sample. In each panel, thick lines show the sample median and the shaded bands enclose the 16–84% and 5–95% ranges. *Top row:* median profiles before (a) and after (b) background subtraction, and the corresponding major- and minor-axis profiles of the raw (c) and background-subtracted (d) data (§3.2). Panels (c) and (d) also include the implied axis ratio $q(r)$ in dark blue (faded within $\theta_{\text{beam}} \approx 7.2$ pc). The median integrated σ_{mom0} is shown as a dotted line. *Bottom row:* the background-subtracted, peak-normalized median profile (e), repeated with the beam-convolved Gaussian (f) and power-law (g) fits of Table 1. Single-component models are shown as solid lines, while two-component models are dashed; the dotted line in (g) marks the best-fit break radius. In panel (h) we also show the corresponding cumulative flux fraction within 60 pc.

The broken power law is formally preferred over the double Gaussian, but ΔBIC ’s systematic range spans zero (Table 1) so the underlying functional form is not strongly constrained. Unresolved substructure flattening the core can push r_{break} toward the beam scale, so it is best read as an upper limit on the transition radius. Were the break primarily set by the beam, however, individually fit peaks would cluster near θ_{beam} rather than spanning 8–28 pc as we observe below.

4.2.2. Core-like inner profiles with a steep fall-off

Assuming a spherical distribution whose integrated intensity traces the column density,⁹ an inverse Abel transform converts the projected profile of Eq. 4 into a three-dimensional density profile with $\rho \propto r^{-\alpha} = r^{-(\gamma+1)}$ in each regime and a transition near $r \approx r_{\text{break}}$. For our data, this implies $\alpha_{\text{in}} = 1.65^{+0.02}_{-0.05}$ and $\alpha_{\text{out}} = 2.65^{+0.05}_{-0.10}$. The double Gaussian deprojects to two three-dimensional Gaussians of unchanged widths; here, a compact $\sigma_{\text{G},1} \approx 7$ pc core embedded in a $\sigma_{\text{G},2} \approx 32$ pc envelope.

⁹ Supported for our data by the radial $^{13}\text{CO}/^{12}\text{CO}$ ratio in Appendix C.

The inner power-law slope falls within the $\alpha \approx 1.5$ –2 predicted for systems dominated by scale-free gravitational collapse (e.g., Girichidis et al. 2014; Li 2018; Vázquez-Semadeni et al. 2019), extending Milky Way observations of this behavior to ~ 10 pc structures. For a spherical power law, it corresponds to a density PDF tail of slope $3/\alpha_{\text{in}} \approx 1.8$, consistent with resolved observations of local clouds (Kainulainen et al. 2014) and within the range Meidt et al. (2025) require to match star formation efficiencies in PHANGS. Density profiles this steep also imply more centrally concentrated clouds than the uniform geometry often assumed for extragalactic virial analyses (Bertoldi & McKee 1992).

The steep outer profile is less expected. Galactic profiles more often flatten outward from core to clump scales (Beuther et al. 2024, 2025) with parent clouds that are shallower than the clumps embedded within them (Peretto et al. 2023). Much of the steepening we observe reflects the elongated shape of the peaks rather than a spherically symmetric decline. At large radii, the median falls to background once fewer than half the pixels in an annulus lie on a structure, so an azimuthally averaged profile is truncated by its minor-axis extent. Fitting to the axis-aligned profiles (Fig. 3d) makes this explicit: $\gamma_{\text{out,maj}} = 1.34^{+0.16}_{-0.14}$ against $\gamma_{\text{out,min}} = 3.3^{+0.6}_{-0.5}$,

Table 1. Functional fits to the median stacked radial profile

Model	Parameter	Value $\pm$ stat.	Sys. range	Bias	BIC	RMS
Gaussian (Eq. 1)	I_G	$1.04^{+0.02}_{-0.01}$	0.86–1.09	...	−140	0.053
	σ_G	$8.9^{+0.3}_{-0.4}$	7.8–9.3	...		
Double Gaussian (Eq. 2)	$I_{G,1}$	$1.01^{+0.03}_{-0.02}$	0.69–1.02	...	−184	0.020
	$I_{G,2}$	$0.13^{+0.03}_{-0.02}$	0.13–0.56	...		
	$\sigma_{G,1}$	$6.9^{+0.2}_{-0.4}$	4.0–6.9	...		
	$\sigma_{G,2}$	32^{+4}_{-3}	11–32	...		
Power law (Eq. 3)	I_P	$0.35^{+0.01}_{-0.01}$	0.29–0.48	...	−145	0.048
	γ	$1.00^{+0.02}_{-0.02}$	0.68–1.15	...		
Broken power law (Eq. 4)	I_P	$0.52^{+0.02}_{-0.02}$	0.41–0.54	+0.01	−208	0.012
	γ_{in}	$0.65^{+0.02}_{-0.05}$	0.60–0.78	−0.01		
	γ_{out}	$1.65^{+0.05}_{-0.10}$	1.43–3.14	−0.02		
	r_{break}	$9.3^{+0.6}_{-1.2}$	8.3–12.5	−0.2		
$\Delta\text{BIC (PL - BPL)}$		63^{+9}_{-11}	56–83			
$\Delta\text{BIC (G - 2G)}$		44^{+8}_{-5}	40–61			
$\Delta\text{BIC (2G - BPL)}$		24^{+6}_{-10}	−21–+40			

NOTE—Fits to the background-subtracted, peak-normalized median stacked profile; models are convolved with the synthesized beam ($\theta_{\text{beam}} \approx 7$ pc) before comparison. Statistical uncertainties are the 16–84% range of 1000 bootstrap resamples of the stack, the systematic range spans six plausible methodological variants of the pipeline, and bias is the median parameter offset recovered from a fiducial model injection test (Appendix B). Lower BIC and RMS point to a better fit; positive ΔBIC strongly favors the two-component models and favors a broken power law over the double Gaussian.

so the profile along each ridge is far shallower than the profile across it. Our background subtraction strongly suppresses the minor axis so that $\gamma_{\text{out},\text{min}}$ may be best read as an upper bound; nonetheless, the elongation persists without it (Fig. 3c; Appendix A). This steep-edged cross-section resembles the radial profiles of Galactic filaments across a range of scales (e.g., Arzoumanian et al. 2011; Zucker et al. 2018). Even so, a milder steepening (with $\Delta\gamma \approx 0.7$) remains along the ridge and suggests that non-spherical geometry is not the full explanation.

Although the exact value of γ_{out} depends on the background treatment, the two-component form with a steep outer profile is not an observational or methodological artifact. In Appendix B, we show that it holds in 100% of bootstrap resamples, across all six methodological variants (including without background subtraction), and in every injected realization. Sensitivity is not a limiting factor, as the noise floor σ_{mom0} lies well below the outer profile (Fig. 3). Nor does blending with neighboring structures create the break: a watershed segmentation assigning each pixel to a single peak (van der Walt et al. 2014) removes most emission beyond 30 pc, steepening γ_{out} to 2.2 and moving r_{break} to 13 pc with γ_{in} nearly unchanged. In fact, emission at large radii does

not drive the steepening, as refitting within $r < 30$ pc returns a *steeper* $\gamma_{\text{out}} = 2.0^{+0.1}_{-0.3}$ at $r_{\text{break}} = 11.1^{+0.5}_{-1.5}$ pc. Finally, our result is not an artifact of stacking. Fit individually, 78% of peaks prefer a broken over a single power law, with $\gamma_{\text{in}} = 0.7^{+0.6}_{-0.3}$ and $\gamma_{\text{out}} = 2.0^{+3.0}_{-1.9}$ at $r_{\text{break}} = 15^{+13}_{-7}$ pc.

We briefly consider two physical effects that may also shape the residual edge. First, CO can be photodissociated in the diffuse outskirts of clouds. At 20–30 pc, raw profile intensities (before background subtraction) of ~ 10 K km s $^{-1}$ imply $N(\text{H}_2) \approx 3 \times 10^{21}$ cm $^{-2}$, within a factor of a few of where CO-dark layers may develop (Wolfire et al. 2010). Second, the edge could mark where self-gravity gives way to an envelope coupled to the galactic potential (Meidt et al. 2018, 2020, 2025) or bounded by the surrounding turbulent medium (Field et al. 2011). Indeed, the peaks’ internal turbulent pressure, $P_{\text{int}}/k_B \sim \rho\sigma_v^2/k_B \sim 10^5$ K cm $^{-3}$, is comparable to the dynamical equilibrium pressure across our field, $P_{\text{DE}}/k_B \approx 1\text{--}3 \times 10^5$ K cm $^{-3}$ (Sun et al. 2020, 2022). Under galactic forcing, isothermal gas should settle into an envelope of width σ_v/κ (Meidt et al. 2025); our median $\sigma_v \approx 4$ km s $^{-1}$ and $\kappa \approx \sqrt{2}V_c/R \approx 100$ km s $^{-1}$ kpc $^{-1}$ (Hlavacek-Larrondo et al. 2011) pre-

dict 40 pc, comparable to the $\sigma_{G,2} \approx 32$ pc envelope of the double Gaussian fit. Local variations in σ_v and κ would naturally produce a (possibly environmentally mediated) range of break radii instead of a universal characteristic scale, as we observe and as O25 found for bound structures in this same field.

4.3. The relation between profile shape and local star formation

We quantify each peak’s association with local H α or 21 μm emission by integrating over a 30 pc aperture around the peak. At 1'' resolution, neither tracer resolves the CO peaks themselves, so these measurements characterize the local star-forming environment of each peak. In Fig. 4, we stack profiles in bins of nearby tracer brightness to explore how the profile shape changes. Table 2 correlates these tracer measurements with per-peak structure metrics.

Brightness—Figure 4 and Table 2 clearly demonstrate that CO peaks near H α and 21 μm emission are systematically more luminous, with brighter central intensities and higher inferred masses. The CO–H α correlation is well established at $\gtrsim 100$ pc (e.g., Pessa et al. 2021), but is expected to weaken at smaller scales as individual regions decorrelate (e.g., Schrubba et al. 2010; Onodera et al. 2010; Kruijssen et al. 2019). Our measurements show that it persists down to 7–30 pc and to individual CO peaks. The correlation with 21 μm is stronger still, extending to smaller scales the finding that 21 μm emission traces a combination of the gas column and dust heating by recent star formation (e.g., Leroy et al. 2023; Belfiore et al. 2023). In Appendix C, we use matched ^{13}CO imaging to show that the observed profile brightness enhancement follows real variation in the gas column density, limiting the contribution of line opacity or CO abundance effects to $\lesssim 0.1$ dex.

Profile shape and concentration—Normalized profiles (Fig. 4, second row) show that this brightness trend is not accompanied by a change in profile shape: median profiles are consistent across both H α and 21 μm bins, with no systematic difference in steepness or width. The absence of a systematic trend is notable because the density profile has been treated as an evolutionary indicator for molecular clumps, expected to steepen as gas collapses (Butler & Tan 2012; Gómez et al. 2021; Li & Zhou 2022; Xu et al. 2024) and to flatten again once HII regions fragment and disperse their surroundings (Dideion et al. 2015; Gieser et al. 2023). These trends are calibrated on $\lesssim 1$ pc core and clump scales, but we find that profile steepness carries little predictive information about star formation at 7–30 pc scales.

Table 2. Rank correlation between profile structure metrics and star formation tracers.

Metric	Tracer	$(\rho_S, p)_{\text{raw}}$	$(\rho_S, p)_{\text{partial}}$
$I_{\text{CO}}(0)$	H α	+0.40, $< 10^{-8}$	...
	21 μm	+0.61, $< 10^{-22}$	...
MCI	H α	+0.13, 0.06	...
	21 μm	−0.03, 0.64	...
Scatter	H α	+0.17, 0.01	+0.24, $< 10^{-3}$
	21 μm	+0.02, 0.79	+0.11, 0.10
Asymmetry	H α	+0.14, 0.03	+0.27, $< 10^{-4}$
	21 μm	+0.05, 0.51	+0.25, $< 10^{-3}$
Elongation	H α	+0.17, 0.01	+0.23, $< 10^{-3}$
	21 μm	+0.05, 0.47	+0.14, 0.05

NOTE—Spearman rank correlation coefficients ρ_S and their corresponding p-values, with partial correlations controlling for central CO brightness. Tracer association is defined by the enclosed flux (H α) or mean surface brightness (21 μm) within 30 pc. Metrics are computed within $r < 30$ pc, as follows (§4.3): $I_{\text{CO}}(0)$ — the background-subtracted peak (central) CO intensity; MCI — the multiple concentration index (Thilker et al. 2022), a measure of peak concentration; Scatter — mean normalized azimuthal scatter (Eq. 5); Asymmetry — normalized, intensity-weighted mean RMS deviation between four axis-aligned quadrant profiles; Elongation — the deconvolved second-moment axis ratio (§3.2).

Similarly, we find that star-forming and quiescent structures have equivalent concentrations on our scales. We quantify profile concentration across the peak population using the multiple concentration index (MCI) of Thilker et al. (2022). The MCI averages several renormalized concentration indices (enclosed-flux ratios $F(r_{\text{in}})/F(r_{\text{out}})$) evaluated at $r = [1, 2, 3, 4] \times \theta_{\text{beam}}$, capturing concentration over a wider radial range than a single index. The MCI rank correlations presented in Table 2 imply that the processes setting the azimuthally averaged radial profile act independently of, or before, stellar feedback having a large effect.

CO-bright peaks are the exception. The third column of Fig. 4 bins profiles by their central intensity $I_{\text{CO}}(0)$ and shows that the brightest peaks are systematically broader and fall off more slowly, in agreement with the results of He et al. (2026) at 150 pc resolution. Such elevated wings could reflect a simple intensity-size relation or these peaks’ location within extended reservoirs of relatively dense CO-bright gas, though opacity effects (explored in Appendix C) may also contribute.

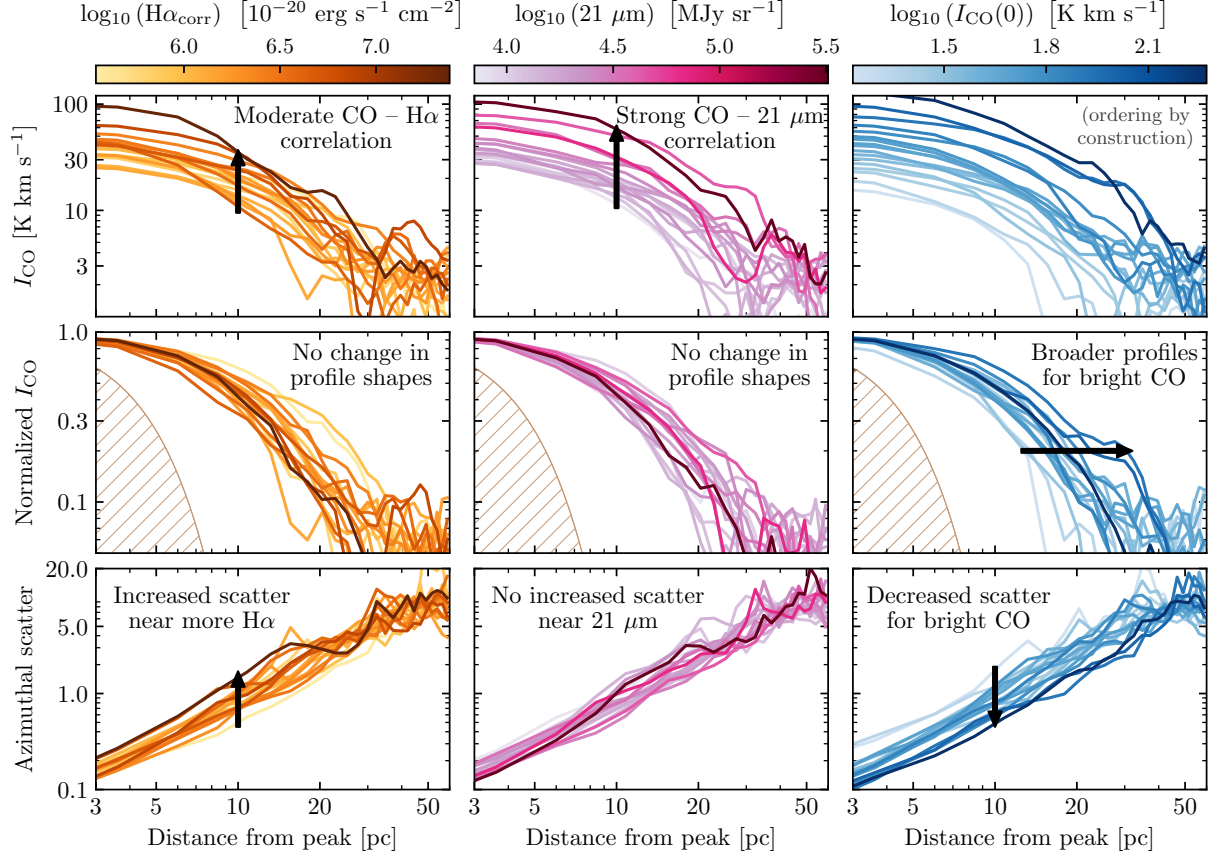


Figure 4. Binned CO(2–1) radial profiles around the peaks. The left and center columns bin profiles by star formation tracer ($H\alpha$ flux or JWST $21\ \mu\text{m}$ surface brightness) measured within 30 pc of each peak. The right column bins instead by the peaks’ own central CO intensity, $I_{\text{CO}}(0)$. Within each column, peaks are sorted into 17 equal-number percentile bins of the tracer, and each curve shows the median profile of the peaks in that bin. Rows show, from top to bottom: the background-subtracted, median azimuthally averaged CO intensity I_{CO} (as in Fig. 3b); the same profiles normalized to their innermost radial bin (as in Fig. 3e); and the azimuthal scatter (Eq. 5) as a function of radius. The hatched region in the middle row marks the beam FWHM. Association with $H\alpha$ and $21\ \mu\text{m}$ emission, tracing more active star formation, corresponds to brighter CO peaks but similar profile shapes; however, the $H\alpha$ association is accompanied by increased azimuthal scatter, while the $21\ \mu\text{m}$ association is not.

Regularity and disruption—While local star formation appears to leave the azimuthally averaged profile shape unchanged, it has a significant effect on profile regularity. In the bottom row of Fig. 4, we show the normalized azimuthal scatter, estimated as

$$(P_{84} - P_{16})/P_{50} \quad (5)$$

for percentile P (Fig. 3). Within $r < 30$ pc, the scatter appears elevated for $H\alpha$ -bright peaks and suppressed for CO-bright peaks; however, no corresponding trend is visible for $21\ \mu\text{m}$.

To quantify profile regularity across the peak population, we compute three scalar metrics within $r < 30$ pc:

1. Scatter: the mean of Eq. 5’s normalized azimuthal scatter;

2. Asymmetry: the mean normalized, intensity-weighted RMS deviation between four radial profiles corresponding to quadrants aligned with the position angle estimate of §3.2;¹⁰
3. Elongation: the deconvolved aspect ratio of intensity-weighted second moments estimated in §3.2.

The resulting correlations with $H\alpha$ and $21\ \mu\text{m}$ association are reported in Table 2. Partial correlations control for each peak’s central CO brightness to isolate the morphological signal. The scatter, asymmetry, and elongation all correlate positively with nearby $H\alpha$ emission, both before and after this control. Individually, these

¹⁰ We prefer this asymmetry metric to the more common index of *Conselice* (2003), which only captures bipolar structure.

correlations are relatively weak ($\rho \approx 0.2\text{--}0.3$), but their consistent sign and the dilution expected from noise, projection, and cluster-gas offsets suggest a real if modest effect. For 21 μm emission, only the asymmetry parameter reaches significance ($p < 0.05$) and only after the partial control. Together, these three metrics point to a coherent morphological signature of disruption that is clearest for H α emission and only marginal for 21 μm .

The H α –21 μm contrast may reflect different phases and evolutionary stages traced by our star formation proxies. Dust heating and the resulting enhancement in 21 μm emission begin when young clusters are still embedded in their natal gas (e.g., [Hassani et al. 2023](#); [Kim et al. 2023](#)), while H α emission is only observable once an HII region is relatively exposed (a transition that can occur rapidly in nearby galaxies; [Hannon et al. 2022](#)). Winds and ionization fronts from exposed HII regions inject energy and momentum into the surrounding ISM, dissociating low-intensity gas and restructuring the local cloud (e.g., [Walch et al. 2012](#); [Tremblin et al. 2014](#); [Didelon et al. 2015](#); [Chevance et al. 2023](#)). Because 21 μm emission arises from dust heated by the ambient radiation field as well as by embedded and exposed young stars, its association with the natal star-forming region, and hence with the disrupted CO phase, may be diluted. If disruption is a hallmark of this exposed phase, scatter, asymmetry, and elongation should rise with H α brightness more closely than with 21 μm , as we observe. Quiescent, CO-rich structures remain relatively undisturbed, while their star-forming neighbors are restructured by feedback from emerging HII regions. Such disruption need not broaden the azimuthally averaged profile, as young clusters are typically displaced from molecular gas peaks by tens of parsecs (e.g., [Ochsendorf et al. 2016](#); [Grasha et al. 2019](#)). Profiles centered on surviving CO maxima can therefore register feedback as asymmetry and scatter without a systematic radial broadening.

5. CONCLUSIONS

We have measured the radial CO profiles of 215 bright peaks at 7 pc resolution in the disk of NGC 253 and compared them to matched MUSE H α and JWST 21 μm imaging.

Peak structure (§4.1 and §4.2)—After background subtraction, the peaks in our catalog have molecular gas masses of $\sim 2 \times 10^5 M_\odot$ within 30 pc, characteristic sizes of ~ 20 pc, and line widths of a few km s^{-1} . Their stacked emission is round near the center but becomes highly elongated in the outskirts, reaching an aspect ratio of 2.4 at 30 pc. Beam-convolved fits to the stacked median require two components: a double Gaussian and

a broken power law both fit well, while single-component models do not. Deprojected into three dimensions, the inner power-law density slope $\alpha_{\text{in}} = 1.65^{+0.02}_{-0.05}$ matches predictions for self-gravitating collapse and literature measurements of Milky Way cores and clumps at $\lesssim 1$ pc, while the profile steepens sharply to $\alpha_{\text{out}} \approx 2.7$ beyond a break at $\lesssim 10$ pc, whose exact radius varies from peak to peak. Much of this steepening is set by the elongated shape of these filamentary structures, while the residual edge along their ridges may reflect a CO-dark layer or a transition to a new kinematic and pressure regime.

Relation to star formation (§4.3)—The presence of nearby H α and 21 μm emission is strongly associated with the CO brightness of individual peaks, showing that these correlations persist to the 30 pc scales that we average over. Notably, the normalized profile shape does not depend on H α or 21 μm brightness, but its regularity does: peaks near bright H α emission show modestly but systematically higher azimuthal scatter, asymmetry, and elongation. For 21 μm -associated peaks, only asymmetry correlates weakly, and only after controlling for the CO intensity. We interpret this contrast as a tentative evolutionary sequence. Feedback associated with the formation of young stars disturbs the gas morphology once HII regions are exposed and bright in H α , while 21 μm emission, coupled to the broader molecular medium over a longer period, dilutes the signal.

Our results provide the first resolved (7–60 pc) view of molecular cloud density structure with matched star formation tracers in the disk of a galaxy beyond the Local Group and indicate that the clearest imprint of star formation on parsec scales lies not in the steepness of the radial profile but in its regularity.

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Facilities: ALMA, JWST, VLT:Yepun

Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), Matplotlib (Hunter 2007), NumPy (Harris et al. 2020), Scipy (Virtanen et al. 2020)

APPENDIX

A. INCLINATION

NGC 253’s relatively high inclination ($i \approx 76^\circ$, McCormick et al. 2013) means that each line of sight traverses a long path through the molecular disk. For the ~ 100 pc FWHM ($\sigma_z \approx 40$ pc) molecular gas scale height estimated for this region by O25 and assuming a simple plane-parallel geometry, sight lines can cross $2\sigma_z \tan i \approx 340$ pc of significant emission. The spectral masking described in §3.1 removes kinematically distinct emission superposed on each peak, but cannot exclude co-moving gas. Here we characterize the residual projection signature on the masked maps empirically.

The elongation of individual peaks in our catalog is weakly but coherently aligned with NGC 253’s disk, consistent with the effects of in-plane projection or suggestive of filamentary structure with a preferential, disk-parallel axis (Zucker et al. 2018; Krahm et al. 2026) or envelopes flattened by anisotropic galactic forcing (Meidt et al. 2025). Comparing each peak’s second-moment position angle (§3.2) to the disk major axis (PA $\approx 51^\circ$, Jarrett et al. 2003), the orientations cluster about the latter with a mean offset of $-7_{-4}^{+4}^\circ$ while a Rayleigh test on the doubled angles (Fisher 1996) confirms they are non-uniformly distributed at high significance ($p_R < 10^{-10}$). This is more significant clustering than found by O25 (their Appendix D) because their dendrogram leaves typically exhibit smaller radii of ~ 10 pc. In §4.2, we show that these inner regions tend to be more symmetrical, partially due to the circular beam. At 30 pc, our median $q \approx 2.4$ begins to approach the thin-layer limit of $\sec i \approx 4.1$, which de-

scribes strongly flattened structures confined to the disk plane.

In Fig. 5 we show the median peak structure over the velocity-masked, background-subtracted moment 0 maps of all 215 peaks, each normalized to its central maximum. When each peak is rotated to align its position angle (highlighting maximum extended structure by construction), a filamentary ridge is visible at 10–20% of the central intensity. Rotated to align with the galaxy position angle instead, the ridge becomes less significant, though still apparent. In the disk frame, $q(30 \text{ pc}) = 1.72_{-0.10}^{+0.11}$ lands well below the peak-frame $q(30 \text{ pc}) = 2.56_{-0.13}^{+0.15}$ but remains significantly elongated. Roughly half of the peak-frame elongation ($q - 1$) can be attributed to a disk-coherent component. Projection thus imposes a modest anisotropy on our sample, increasingly significant toward the faint outskirts of profiles. Because all peaks share a global inclination, this effect cannot drive the differential trends or correlations with H α and 21 μm association that we present in §4.3. Our velocity masking is a reasonably effective, though necessarily incomplete, control for the effects of a highly inclined view.

B. UNCERTAINTIES ON FIT PARAMETERS

Parameter errors taken from the covariance matrix of an unweighted least-squares fit to the stacked profile fail to realistically capture our uncertainties, since they treat the radial bins as fully independent and the peak sample as fixed. In this appendix, we rigorously characterize the uncertainty budget of the stacked-profile fits with three

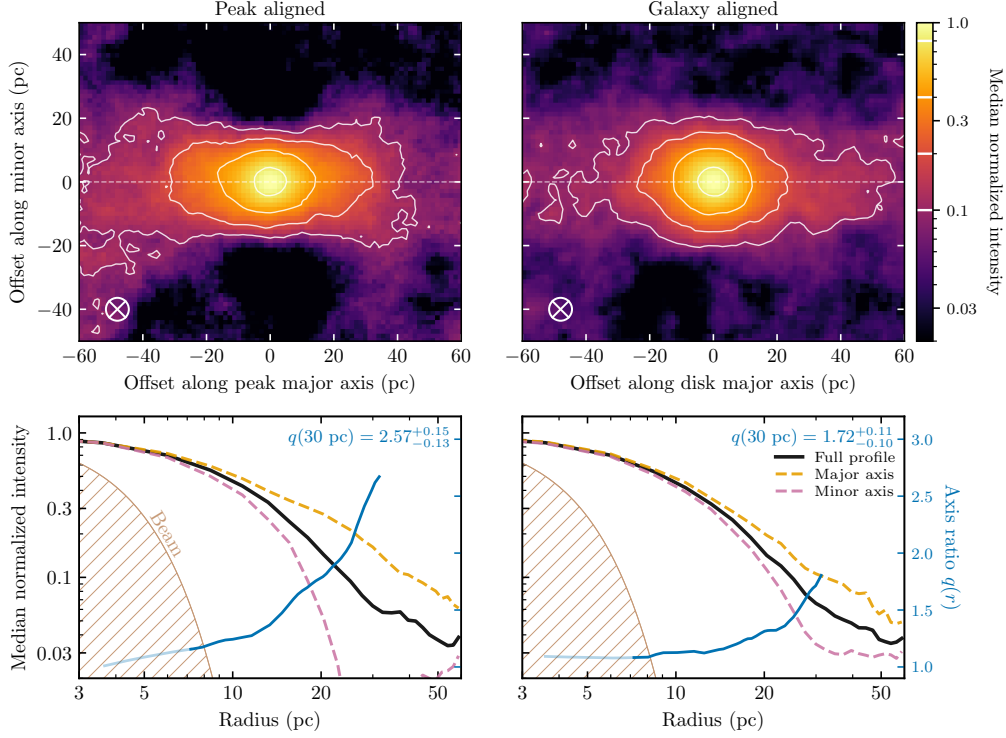


Figure 5. Stacked profiles rotated to align with each peak’s position angle (left column) or to NGC 253’s major axis (right column). **Top row:** Median stacks of the velocity-masked, background-subtracted, peak-normalized moment 0 cutouts of all 215 CO peaks. Contours are shown at 0.1, 0.2, 0.4, and 0.8 times the central intensity, with the beam FWHM indicated. **Bottom row:** Median radial profiles of the stack over the full azimuthal range as well as the major and minor axes, with the axis ratio within 30 pc plotted as in Fig. 3.

tests, collected in Table 1: the statistical and sampling uncertainty, systematic variation under our methodological choices, and the recovery bias of the pipeline itself.

We measure the statistical uncertainty with a non-parametric bootstrap over the peak catalog, drawing 1000 resamples of the 215 peaks with replacement and re-stacking and refitting each realization using the pipeline of §4.2. This approach propagates bin-to-bin correlations and per-peak noise into the distributions. Every fit converges successfully with no parameter at its bound. The resulting asymmetric uncertainties (16–84% range) on our fit parameters are comparable in magnitude to the covariance errors, but the underlying distributions are mildly skewed toward shallower slopes and smaller break radii.

To assess the systematic impact of a range of plausible methodological choices, we repeat the analysis under the following six pipeline variants, including the fiducial: replacing the sigma-clipped background estimator with a flat 10th percentile of each profile, or with its median at $r > 30$ pc; omitting background subtraction entirely and instead fitting models with a free constant; stacking before, rather than after, normalization; and estimating and subtracting the background from the stacked profile

rather than peak-by-peak. Table 1 lists the full range of each parameter across the variations. These ranges should not be combined with the statistical uncertainty *per se*, but rather serve as a conservative test of methodological independence in our qualitative description of profile geometries.

Finally, we quantify the bias imposed by the stacking and fitting procedure by injecting the best-fit broken power law and evaluating it through the same forward model as §4.2. We construct 200 realizations, each stacking 215 synthetic profiles built by adding residuals drawn from profiles in the observed stack, thereby preserving realistic peak-to-peak diversity and noise. Refitting the synthetic stacks recovers the input parameters to within $\lesssim 35\%$ of the statistical uncertainty in all cases, which validates our approach to stacking and fitting the data. To ensure that the two-component structure is not spurious, we also inject the best-fit *single* power law ($\gamma = 1.0$) and build 200 realizations by the same procedure. The injected index is recovered exactly ($\bar{\gamma} = 1.004$), while no realization reproduces a false break in terms of ΔBIC or $\gamma_{\text{out}} - \gamma_{\text{in}}$.

C. OPTICAL DEPTH AND ABUNDANCE EFFECTS

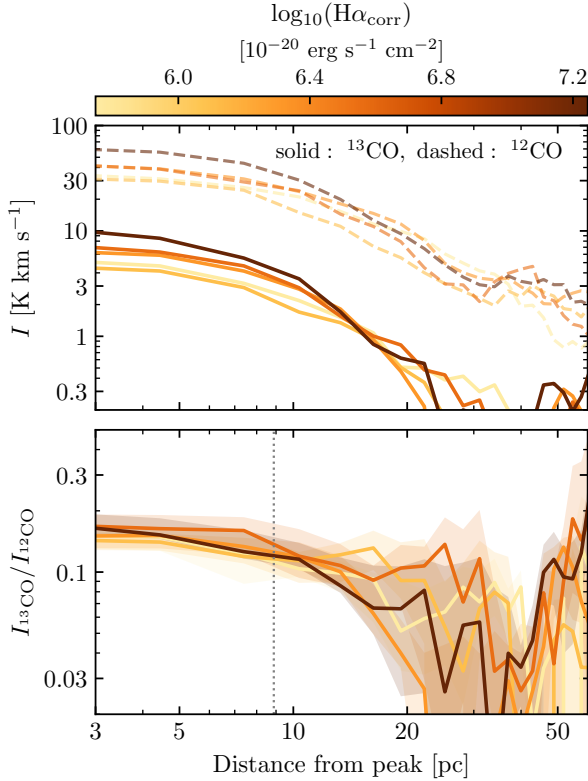


Figure 6. $^{13}\text{CO}(2-1)$ test of whether line opacity or abundance drives the CO–star formation tracer trend. **Top:** Stacked, background-subtracted radial profiles of ^{13}CO (solid) and ^{12}CO (dashed, convolved to $\theta_{\text{beam},^{13}\text{CO}}$) for the 168 peaks with central ^{13}CO S/N ≥ 3 , in equal-count quintiles of H α flux measured within 30 pc (as in Fig. 4). **Bottom:** The $^{13}\text{CO}/^{12}\text{CO}$ ratio of the stacked profiles, with 16–84% bootstrap bands; the dotted line marks the 8.9 pc ^{13}CO beam FWHM. Peaks near brighter H α emission are enhanced in the optically thinner ^{13}CO line as strongly as in ^{12}CO , and the central ratio shows no depression toward the H α -bright bins, indicating that the observed CO(2–1) enhancement reflects gas column density rather than increased saturation of the optically thick ^{12}CO line or a reduced ^{13}CO abundance.

The trends described in §4.2 and §4.3 are measured in CO(2–1) intensity; as rank correlations and profile statistics, they stand irrespective of physical interpretation. Whether brighter CO peaks contain correspondingly more molecular gas, however, depends on two effects, both sensitive to the hard radiation fields near sites of active star formation (Schinnerer & Leroy 2024): (i) the line ratio R_{21} and the conversion factor X_{CO} , which set the CO(2–1)-to-mass scaling; and (ii) the optical depth and CO abundance, which set the ^{12}CO in-

tensity at fixed column. We bound the former from our calibration and the literature, and test the latter directly.

Within nearby galaxy disks, $R_{21} \approx 0.5$ – 0.8 and rises weakly with local star formation activity, though no extragalactic measurement resolves it below ~ 100 pc (den Brok et al. 2021; Leroy et al. 2022; den Brok et al. 2025). In the Milky Way, only strongly heated gas approaches $R_{21} \sim 1$ (Sakamoto et al. 1997; Nishimura et al. 2015), and the shielded phase traced by CO is itself relatively insensitive to radiation field strength (Peñaloza et al. 2018). Our adopted $R_{21} = 0.615$ is calibrated against the best available CO(1–0) data for this field (Sorai et al. 2000; Kuno et al. 2007; O25), which (though unresolved) accounts for the large-scale impact of the starburst environment. Peak-to-peak deviations could bias our trends by at most ~ 0.2 dex if star formation-bright peaks were fully thermalized, and $\lesssim 0.1$ dex over the observed extragalactic range within disks. Furthermore, for $\alpha_{\text{CO}} \propto n^{1/2} T_{\text{ex}}^{-1}$ (Bolatto et al. 2013), elevated temperatures near star formation should offset increases in α_{CO} from higher densities and larger CO-dark envelopes. Dust-based measurements find α_{CO} roughly flat with Σ_{SFR} outside galaxy centers (Sandstrom et al. 2013), while resolved Galactic clouds show factor of a few variations with the column and radiation field. These variations are concentrated in CO-dark envelopes over shielded interiors (Lee et al. 2014; Kong et al. 2015), a bias that would suppress and not enhance CO brightness near heated peaks.

Matched $^{13}\text{CO}(2-1)$ observations from the same ALMA program test the second effect directly. $^{13}\text{CO}(2-1)$ is optically thinner than the ^{12}CO line, and its emissivity per molecule varies little over $T \approx 10$ – 40 K, so it traces the gas column where ^{12}CO saturates. Using the same peak positions, velocity masks, and stacking procedure as for ^{12}CO (§3), we construct background-subtracted ^{13}CO profiles for the 168/215 peaks whose central ^{13}CO profile bin is detected at $\geq 3\sigma$, stack them in H α flux quintiles, and measure their corresponding ratios (Fig. 6). If the observed CO–star formation tracer trend is primarily driven by the gas column density, we would expect ^{13}CO to be enhanced like ^{12}CO , giving a flat or slightly rising ratio with H α brightness. If the correlation is instead driven by increased saturation or by a reduced ^{13}CO abundance from selective photodissociation and suppressed fractionation, the ratio should fall near warm, irradiated gas.

Neither signature is evident in Fig. 6. After convolving to a common $0''.52$ resolution, ^{13}CO correlates with H α emission as strongly as does ^{12}CO ($\rho_S = 0.39$ vs. 0.40) and the variation of most to least associated bins is iden-

tical for the two lines. The stacked ratio also appears mostly flat across intensity bins within ~ 20 pc, beyond which the ^{13}CO data drop to a noisy background. Controlling for ^{12}CO brightness, the ratio– $\text{H}\alpha$ correlation remains mildly positive ($\rho_S = 0.15$, $p = 0.06$), consistent with opacity growing with the column density rather than ^{12}CO brightening at fixed column. The $21\ \mu\text{m}$ association, not shown here for space, behaves similarly at even higher significance. Were the enhancement pro-

duced entirely at fixed column, the ratio would fall in proportion; the measured contrast between the brightest and faintest $\text{H}\alpha$ bins is instead consistent with zero, bounding any such contribution to $\lesssim 0.1$ dex (1σ). We conclude that the CO brightness excess near star formation tracers is a real reflection of higher molecular gas column densities, although remaining systematics can only be removed by matched-resolution CO(1–0) imaging.

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